

SQL Schema Design

Objective

Create a DDL file describing the database schema for a restaurant database. Create the structure for this task.

Requirements

1. Create the `content` Schema

All tables must live inside a dedicated `content` schema — **not** the default `public` schema.

2. Design and Create the Tables

- Analyze the planned database structure.
- Create a `.ddl` file describing **all** tables.
- Choose appropriate data types for each column.
- Use **UUID** as the primary key type for every table.
- Define **foreign keys** for all relationships between tables.
- All tables must be prefixed with the `content.` schema (e.g., `content.database_name`).

3. Define Indexes and Uniqueness Constraints

- Decide which columns need indexes for query performance.
- Decide which columns or column groups require **uniqueness constraints**.
- Create the indexes explicitly in the DDL file.

EXPLAIN ANALYZE: Compare Query Performance With and Without Indexes

One of the key goals of this assignment is to **demonstrate the impact of indexes** on query performance. You must compare query execution **before and after** creating each index.

Workflow

Step 1 — Run the query WITHOUT the index:

```
EXPLAIN ANALYZE
SELECT *
FROM content.database_name
WHERE creation_date = '2020-01-01';
```

Record the output (execution time, scan type — likely `Seq Scan`).

Step 2 — Create the index:

```
CREATE INDEX database_name_creation_date_idx ON content.database_name (creation_date);
```

Step 3 — Run the SAME query WITH the index:

```
EXPLAIN ANALYZE
SELECT *
FROM content.database_name
WHERE creation_date = '2020-01-01';
```

Record the output (execution time, scan type — should now show `Index Scan` or `Bitmap Index Scan`).

Step 4 — Document the comparison.

For each index you create, include a brief note in your submission showing the before/after `EXPLAIN ANALYZE` output and a one-line conclusion (e.g., "Seq Scan 12.5ms → Index Scan 0.8ms").

What to Look For in EXPLAIN ANALYZE Output

Field	Meaning
<code>Seq Scan</code>	Full table scan — reads every row. Slow on large tables.
<code>Index Scan</code>	Uses the index to jump directly to matching rows. Fast.
<code>Bitmap Index Scan</code>	Builds a bitmap from the index, then fetches matching rows. Good for medium selectivity.
<code>actual time</code>	Real execution time in milliseconds.
<code>rows</code>	Number of rows returned or scanned.
<code>Planning Time</code>	Time spent building the query plan.
<code>Execution Time</code>	Time spent actually running the query.

MUST: To get meaningful results, populate your tables with test data (at least a few thousand rows) before running EXPLAIN ANALYZE. On an empty table, every query is fast and the comparison is meaningless.

Acceptance Criteria

1. The repository contains a `.ddl` file with all SQL statements needed to create the database from scratch.
2. Table and column names match the schema from the first lesson.

3. Indexes and uniqueness constraints are defined.
4. All tables are in the `content` schema (not `public`).
5. All primary keys use `UUID`.
6. Foreign keys are defined for all relationships.
7. EXPLAIN ANALYZE comparisons are documented for at least the key indexes, showing before/after performance.

You should use Docker for this assignment. Consider the following commands:

```
docker run -d \
  --name postgres \
  -p 5432:5432 \
  -v $HOME/postgresql/data:/var/lib/postgresql/data \
  -e POSTGRES_PASSWORD=123qwe \
  -e POSTGRES_USER=app \
  -e POSTGRES_DB=movies_database \
  postgres:16
```

Make sure to understand what this command does. Explain it line by line.

```
psql -h 127.0.0.1 -U app -d movies_database
sudo apt install postgresql
docker exec -it postgres psql -h 127.0.0.1 -U app -d movies_database
```

Submission

Place your solution in the `schema_design/` folder of your repository. Include:

- `restaurant_database.ddl` — the complete DDL file
- `explain_analyze_results.md` — your before/after comparison notes

Next steps

- Create docker compose to integrate the flask app and the database.